

LIFE SCIENCES

ANCIENT HUMAN RELATIVES

OUR JOURNEY THROUGH TIME



1

AUSTRALO-
PITHECINES



2

HOMO
HABILIS



3

HOMO
ERECTUS



4

HOMO
HEIDELBERGENSIS



5

NEANDERTHALS



6

DENISOVANS



7

HOMO
FLORESIENSIS



8

HOMO
SAPIENS



DIFFERENT PATHS.
ONE HUMAN STORY.



FOSSILS
TELL STORIES



TOOLS
SHOW
INNOVATION



LIFESTYLES
SHAPED BY
ENVIRONMENTS



GENETICS
REVEALS
CONNECTIONS



ONE PLANET
MANY
JOURNEYS

Ancient Human Relatives

Researching the Human Family Tree

Big question

Is human evolution best understood as a straight line
or as a branching tree?

Name: _____

Class: _____

Date: _____

This workbook will help you research ancient human relatives, compare evidence, and explain why human evolution is more complex than a simple ladder.

Contents

Introduction.....	0	Task 9: Interpret the evidence.....	11
Learning Objectives and Key Vocabulary	1	Explain your choice.....	11
By the end of this project, you should be able to:	1	Planning Your Final Answer.....	12
Key vocabulary	1	Planning table	12
Misconceptions Starter	1	Useful structure	12
Task 1: True, false or partly true?.....	1	Final Written Answer	13
Reflection question.....	1	Page 17 - Self-Assessment	14
Research Sources	2	Task 10: Check your work.....	14
Source reliability checklist	2	Best part of my work	14
The Main Groups	2	One thing I could improve	14
Extension option.....	2	Page 18 - Peer Review	15
Choosing Your Research Group	3	Page 19 - Teacher Feedback	16
Task 2: Choose one group to research in detail.....	3	Strengths	16
First impressions	3	Target	16
Research Table: Basic Facts	4	Optional Extension - Homo naledi.....	17
Evidence box	4	Extension task.....	17
Anatomy and Adaptations.....	5	Extension written question	17
Task 3: Body structure.....	5	Teacher Notes and Source Links ... Error! Bookmark not defined.	
Analysis question	5	Suggested source links Error! Bookmark not defined.	
Lifestyle and Behaviour.....	6		
Task 4: How did they live?.....	6		
Careful wording task	6		
Why This Group Matters.....	7		
Task 5: Importance in human evolution.....	7		
Paragraph	7		
Debate and Uncertainty.....	8		
Task 6: What are scientists unsure about?.....	8		
Timeline Task	9		
Task 7: When did they live?.....	9		
Timeline space	9		
Timeline question	9		
Comparison Task	10		
Task 8: Compare three groups	10		
Comparison questions.....	10		
Evolution: Ladder or Tree?	11		

Introduction

Human evolution is often shown incorrectly as a straight line:

ape -> early human -> modern human

This is misleading. Humans did not evolve in a simple chain. Instead, human evolution was more like a branching tree, with different human relatives living at different times, sometimes overlapping, sometimes spreading to new areas, and sometimes dying out.

In this project, you will research several ancient human relatives and use evidence to explain how they fit into the human evolutionary story.

Key idea

Modern humans, *Homo sapiens*, are the only surviving human species today. However, for much of the past few million years, Earth was home to more than one type of human or human-like species.

Your task is to investigate this diversity.

Learning Objectives and Key Vocabulary

By the end of this project, you should be able to:

1. Describe key features of several ancient human relatives.
2. Explain why human evolution is not a simple ladder.
3. Use fossil, archaeological and genetic evidence to compare different hominin groups.
4. Identify where there is uncertainty or debate in the evidence.
5. Produce a clear written explanation using evidence from at least three groups.

Key vocabulary

Misconceptions Starter

Task 1: True, false or partly true?

For each statement, decide whether it is true, false or partly true. Then explain your answer.

Statement	True / False / Partly true	Explanation
Humans evolved from monkeys.		
Evolution always makes organisms better.		
Human evolution was a straight line leading to us.		
Neanderthals were unintelligent cave people.		
Bigger brains always mean a species is more successful.		
Homo sapiens lived at the same time as other human species.		

Reflection question

Which misconception do you think is the most common, and why?

Research Sources

Use the recommended research sources below. You may use other sources, but they must be reliable.

Source	Best used for
Smithsonian Human Origins	Species profiles, timelines, fossils, key features
Natural History Museum	Accessible explanations and recent discoveries
Australian Museum Human Evolution	Pupil-friendly summaries of human evolution
Smithsonian Human Evolution Interactive Timeline	Comparing when different groups lived

Source reliability checklist

Have you thought about **each source** to decide if you think it is a reliable source of information?

Question	Yes / No
Is it from a museum, university, scientific organisation or trusted educational source?	
Does it give evidence, not just opinions?	
Does it explain uncertainty or debate?	
Is it written clearly enough for you to understand?	
Does it avoid exaggerated claims?	

The Main Groups

In this project, you will investigate these ancient human relatives:

Group	Main reason for studying it
Australopithecus afarensis	Early bipedalism; Lucy
Homo habilis	Early Homo; tool-use debate
Homo erectus	Major migration out of Africa
Homo heidelbergensis	Possible ancestor of later human groups
Neanderthals	Close relatives of modern humans
Denisovans	Known mainly from DNA evidence
Homo floresiensis	Small island human species
Homo sapiens	Modern humans; comparison group

Extension option

Group	Why it is interesting
Homo naledi	Strange mixture of primitive and modern traits; debated behaviour

Choosing Your Research Group

Task 2: Choose one group to research in detail

My chosen group is:

I chose this group because:

First impressions

Before researching, what do you already know or think you know about this group?

Research Table: Basic Facts

Complete this table for your chosen group.

Question	Answer
Scientific name / group name	
Approximate date range	
Main fossil locations	
Continent or region	
Was it definitely a member of the genus Homo?	
Did it overlap with Homo sapiens?	
Is it extinct?	

Evidence box

What is the strongest evidence we have for this group? Tick any that apply.

Type of evidence	Tick
Fossils	
Stone tools	
Footprints	
DNA	
Cave art or symbolic objects	
Burial evidence	
Other archaeological evidence	

Explain your answer:

Anatomy and Adaptations

Task 3: Body structure

Feature	Description
Approximate height	
Brain size or skull shape	
Jaw / teeth	
Arms and legs	
Walking ability	
Strength or body build	
Other distinctive features	

Analysis question

Which physical feature do you think was most important for this group's survival?

Feature:

Why was it useful?

Lifestyle and Behaviour

Task 4: How did they live?

Question	Answer
What did they probably eat?	
Did they use tools?	
Did they use fire?	
Did they hunt, scavenge or gather food?	
Did they live in groups?	
Is there evidence of communication or culture?	
Is there evidence of caring for others?	

Careful wording task

Some behaviours are hard to prove from fossils. Complete these sentence starters using cautious scientific language.

The evidence suggests that...

It is possible that...

Scientists are uncertain whether...

Why This Group Matters

Task 5: Importance in human evolution

Explain why your chosen group is important. You could write about:

- bipedalism
- tool use
- migration
- brain size
- survival in different environments
- relationship to Homo sapiens
- interbreeding
- extinction
- scientific debate

Paragraph

My chosen group is important in human evolution because...

Debate and Uncertainty

Human evolution is difficult to study because the evidence is incomplete. Fossils are rare, DNA does not always survive, and scientists may disagree about how to classify fossils.

Task 6: What are scientists unsure about?

Find one debate or uncertainty about your chosen group.

Question	Answer
What is the debate?	
Why is it difficult to answer?	
What evidence would help solve it?	
What is your own judgement?	
Useful phrases The evidence suggests... / However, this is uncertain because... / Some scientists argue... / This is difficult to prove because... / A stronger conclusion would require...	

Timeline Task

Task 7: When did they live?

Place the groups below on a timeline.

Group	Approximate date range
Australopithecus afarensis	
Homo habilis	
Homo erectus	
Homo heidelbergensis	
Neanderthals	
Denisovans	
Homo floresiensis	
Homo sapiens	

Timeline space

Draw your timeline and stick it in here. Include dates, overlap between groups, labels, and a key if needed.

Timeline question

Which groups overlapped with Homo sapiens?

What does this suggest about human evolution?

Comparison Task

Task 8: Compare three groups

Choose your own group, Homo sapiens, and one other ancient human relative.

Feature	My chosen group	Homo sapiens	Third group
Date range			
Location			
Brain / skull			
Body shape			
Tools / behaviour			
Evidence available			
Main uncertainty			

Comparison questions

Which two groups are most similar? Why?

Which two groups are most different? Why?

Evolution: Ladder or Tree?

Task 9: Interpret the evidence

Look back at your timeline and comparison table. Which model is more accurate?

Model	Tick
Human evolution was a straight ladder.	
Human evolution was a branching tree.	

Explain your choice

Use evidence from at least three groups.

Human evolution is better described as a _____ because...

Planning Your Final Answer

Final question

Human evolution is best understood as a branching tree, not a straight ladder. Evaluate this statement using evidence from at least three ancient human relatives.

Planning table

Point	Evidence	Explanation
Point 1		
Point 2		
Point 3		
Counterpoint / uncertainty		
Final judgement		

Useful structure

Paragraph 1: Introduce the idea that human evolution is often misunderstood.

Paragraph 2: Use evidence from one early group, such as *Australopithecus afarensis*, *Homo habilis* or *Homo erectus*.

Paragraph 3: Use evidence from later groups, such as Neanderthals, Denisovans or *Homo floresiensis*.

Paragraph 4: Explain overlap, interbreeding or uncertainty.

Conclusion: Decide whether the tree model is better than the ladder model.

Final Written Answer

Final question

Human evolution is best understood as a branching tree, not a straight ladder. Evaluate this statement using evidence from at least three ancient human relatives.

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Self-Assessment

Task 10: Check your work

Success criterion	Tick
I described at least three ancient human relatives.	
I included accurate dates or approximate time periods.	
I included evidence from fossils, tools, DNA or archaeology.	
I explained why human evolution is not a straight line.	
I included at least one uncertainty or debate.	
I used scientific vocabulary correctly.	
I wrote a clear final judgement.	

Best part of my work

One thing I could improve

Peer Review

Swap your work with another pupil.

Peer reviewer name:

Question	Peer response
What is one strong part of this work?	
Is there enough evidence?	
Is the explanation clear?	
Has the student explained uncertainty or debate?	
What should they improve?	

Teacher Feedback

Strengths

Target

Optional Extension - Homo naledi

Extension task

Research Homo naledi. This species is especially interesting because it had a mixture of primitive and more human-like features. Scientists disagree about how to interpret some of the evidence linked to it.

Question	Answer
Where was Homo naledi discovered?	
When did it live?	
What features were primitive?	
What features were more human-like?	
What behaviours have been suggested?	
Why are these behaviours debated?	

Extension written question

Why does Homo naledi make human evolution harder to understand?
